

Avogadro's law of combining volumes

Learning outcomes

By the end of this section you should be able to:

- Describe the factors that affect the number of particles of gas in a sealed container.
- State Avogadro's law that there are equal volumes of all gases at constant temperature and pressure conditions.
- Solve problems using the mole ratio for gaseous reactants and products in reaction equations.

So far, you have learned how to calculate the number of elementary entities for solids and aqueous solutions. What about for gases? Recall in [section S1.1.1a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/>) that gas particles are very far apart from each other and that the container they are held in is mostly empty space. How can we determine the number of elementary entities that are inside? As these particles are constantly in motion, it would be very difficult to count them, even if we could see them. We must make several measurements that affect the number of particles that can fit inside the container while still allowing the substance inside to behave as a gas.

Factors affecting gases

When a sample of gas is sealed inside a closed container, the particles are constantly in motion because they have kinetic energy. If one particle hits another particle or the sides of the container, it bounces off in another direction with no change to the amount of energy it has. The number of gas particles that can fit inside a closed container is affected by a few different variables:

- The volume of the container – the larger the container, the more particles can fit.
- The temperature of the gas – greater temperatures mean more kinetic energy and therefore greater speeds the particles will travel. At low temperatures, the particles move slowly, so more

particles can fit inside the container. At high temperatures, the particles need more space to move, therefore fewer particles can be accommodated.

- The pressure of the gas. Pressure is a measurement of the frequency of the collisions of the gas particles with the sides of the container. At low pressures, there are fewer collisions with the sides of the container because there are fewer particles present. When more particles are present, the pressure will be higher.

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Pressure and temperature affect the volume a gas will occupy. These two conditions can vary greatly throughout the day in a single location and drastically in different cities around the world. When scientists conduct experiments involving gases, temperature and pressure conditions are carefully controlled and monitored so that these experiments can be reproduced with consistent results. There are also agreed upon standard temperature and pressure conditions that scientists use for consistency with experimental conditions. You will learn more about these in section S1.5.3 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/molar-volume-of-an-ideal-gas-and-gas-laws-id-44352/>).

These three variables together determine the number of gaseous elementary particles that can fit in a container. Since the container is mostly empty space and the gas particles have negligible volumes, an equal number of gas particles will occupy the same volume measured under the same conditions of temperature and pressure. This is known as Avogadro's law.

In **Figure 1**, the balloons all contain different gases at the same temperature and pressure and occupy the same volume. What do you notice about the number of particles inside?

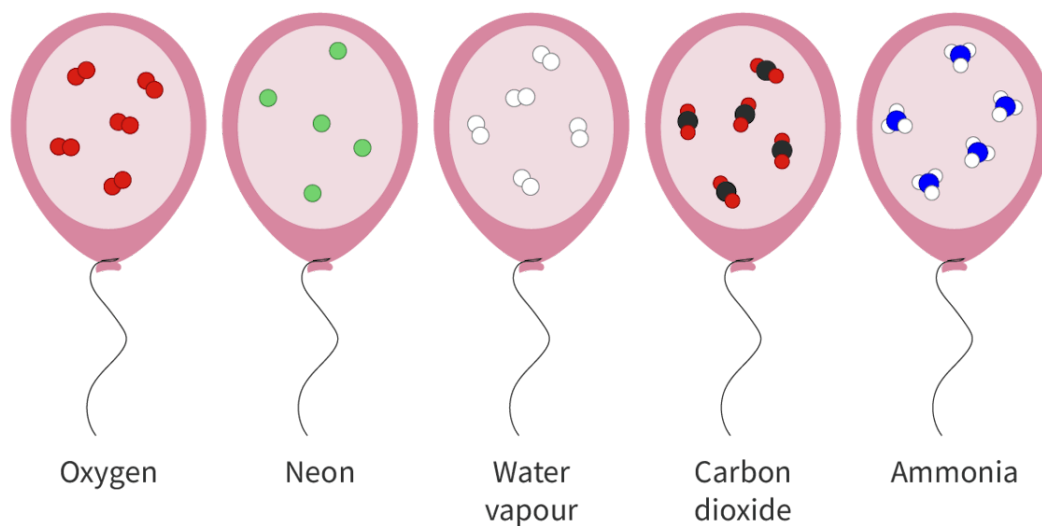


Figure 1. The balloons contain different gases but all have the same volume and contain the same amount of particles.

There are the same number of gas particles at constant temperature, pressure and volume. Do all of these balloons have the same mass? What would affect the mass that each of these balloons have?

No, these balloons do not have the same mass, since different atoms have different masses. We must use the molar mass and the amount, in mol, to determine the mass of each balloon.

Molar masses for different gases.

Balloon	Gas	Molar mass
A	O ₂	32
B	Ne	20
C	H ₂ O	18
D	CO ₂	44
E	NH ₃	17

Therefore, balloon D would have the greatest mass and balloon E would have the lowest mass.

Arrange these balloons **Interactive 1** from smallest to greatest mass.

Smallest mass

Mid mass

Largest mass

Ar

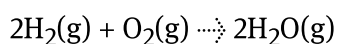
F₂

O₂

☒ Check

Interactive 1. Ordering Gases by Mass.

Avogadro's law is helpful for determining the quantity of gases involved in chemical reactions. For example, water vapour can be made from hydrogen and oxygen gases according to the reaction below:



Using Avogadro's law, we can assume that under constant temperature and pressure conditions, twice the volume of hydrogen gas is required to react with oxygen gas and that the quantity of water vapour produced will be equal to the volume of hydrogen gas that reacted and twice the volume of oxygen gas that reacted. For example, if 1.00 dm³ of hydrogen gas and 0.50 dm³ of oxygen gas reacted, then 1.00 dm³ of water vapour will be produced (**Figure 2**).

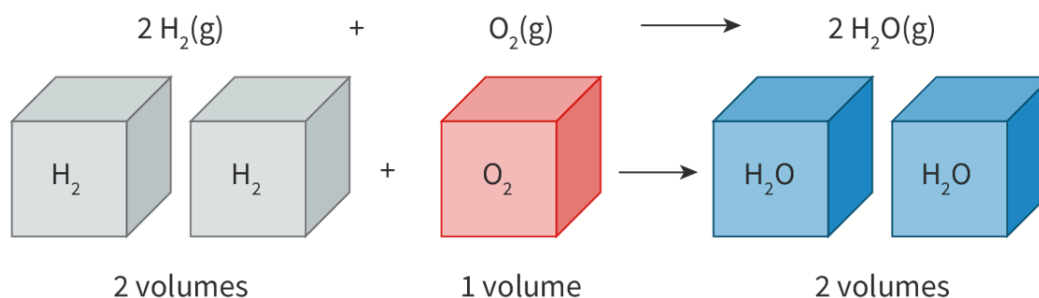


Figure 2. The molar volume of gases involved in a chemical reaction can be predicted using Avogadro's law.

 More information for figure 2

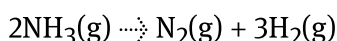
The diagram depicts Avogadro's law using a visual representation of gas volumes in a chemical reaction. It shows two gray cubes labeled H_2 (representing hydrogen gas) with the text "2 volumes" beneath them, indicating two volumes of hydrogen gas. Next to them is a red cube labeled O_2 (for oxygen gas) with "1 volume" beneath it, representing one volume of oxygen gas. An arrow points rightward from the hydrogen and oxygen cubes, leading to two blue cubes labeled H_2O (representing water vapor) with the text "2 volumes" beneath, indicating the formation of two volumes of water vapor. The layout visually supports the explanation that twice the volume of hydrogen gas combines with one volume of oxygen gas to produce a volume of water vapor equal to the hydrogen gas's volume, adhering to Avogadro's law.

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This law will become very helpful to you when you learn more about chemical reactions in [section R2.1.1 \(https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/chemical-equations-id-43480/\)](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/chemical-equations-id-43480/).

Worked example 1

Determine the volumes of hydrogen and nitrogen gases produced when 5.00 dm^3 of gaseous ammonia decomposes under constant temperature and pressure according to the reaction equation below:



Solution steps	Calculations
Step 1: use Avogadro's law for N ₂ and NH ₃ .	1 volume of N ₂ (g) is produced from 2 volumes of NH ₃ (g) that decompose. 2 NH ₃ (g) : 1 N ₂ (g) 5.00 dm ³ NH ₃ (g) :
Step 2: use Avogadro's law for H ₂ and NH ₃ .	3 volume of H ₂ (g) is produced from 2 volumes of NH ₃ (g) that decompose. 2 NH ₃ (g) : 3 H ₂ (g) 5.00 dm ³ NH ₃ (g) :
Step 3: state the answer with appropriate units and the number of significant figures used in rounding.	So 2.50 dm ³ N ₂ (g) and 3.75 dm ³ H ₂ (g) are produced when 5.00 dm ³ NH ₃ (g) decomposes.

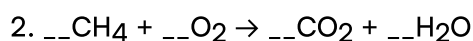
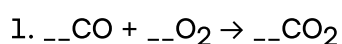
Study skills

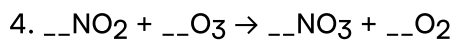
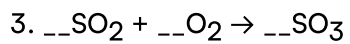
Avogadro's law only applies to gaseous substances in a reaction equation. Be sure to check that you are only applying the ratio to species in the gaseous state and not to substances that are solids, liquids or aqueous solutions.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Applying key ideas and facts in new contexts
- **Time required to complete activity:** 15–20 minutes
- **Activity type:** Individual activity

Balance the reaction equations and model the chemical reactions using cubes to model 1 dm³ units.

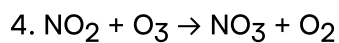
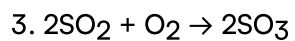
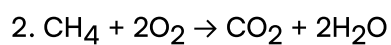
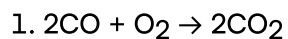




Using your answers above, complete the chart below with the quantities of each gas involved in the chemical reaction:

CO

O₂CO₂1.00 dm³CH₄O₂CO₂H₂O2.00 dm³SO₂O₂SO₃2.50 dm³NO₂O₃NO₃O₂

10.0 dm³

Using your answers above, complete the chart below with the quantities of each gas involved in the chemical reaction:

CO

O₂CO₂1.00 dm³0.500 dm³1.00 dm³CH₄O₂CO₂H₂O1.00 dm³2.00 dm³1.00 dm³2.00 dm³SO₂O₂SO₃

2.50 dm^3 1.25 dm^3 2.50 dm^3 NO_2 O_3 NO_3 O_2 10.0 dm^3 10.0 dm^3 10.0 dm^3 10.0 dm^3